

Digital Electronics & Pulse Technique

Operational Amplifier (OP- Amp)

[high gain differential amplifier with high I/P impedance & low O/P impedance]

Intro :

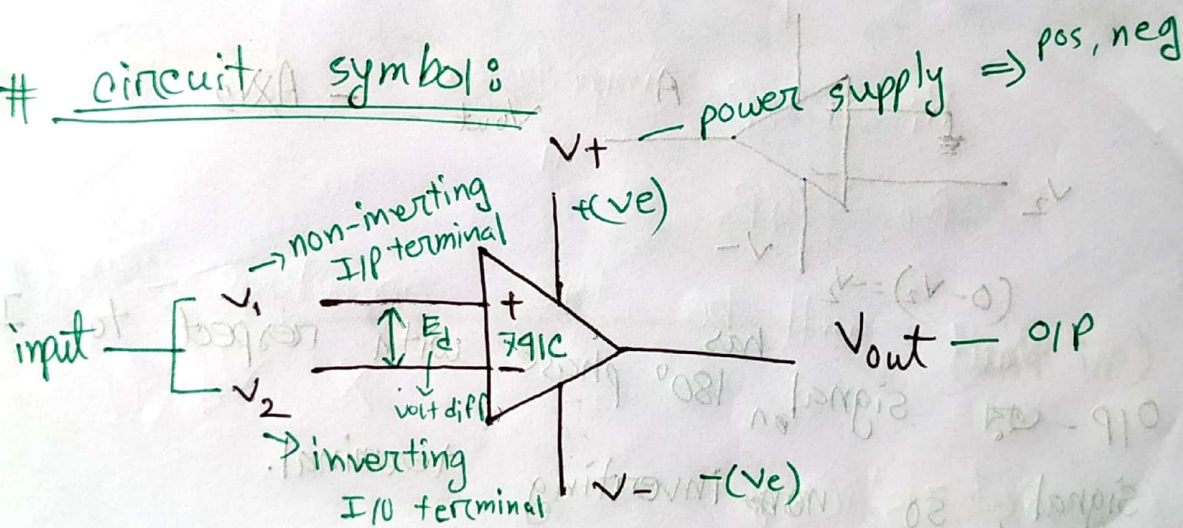
- amplify input signal

differentiation, integration

any called op-amp?

mathematical operation (Add, sub, etc) করার জন্য use করা হতো তাই operational amp বলে

circuit symbol :

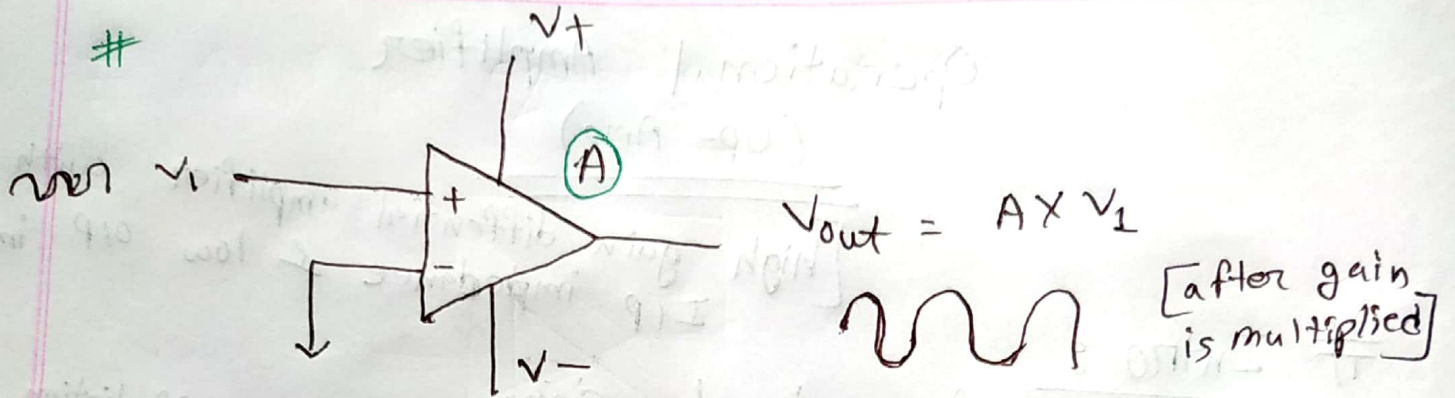


this amplifier amplify করে ২টি input-এর difference-কে,

V_1, V_2 যদি input, আর A, gain of op-amp

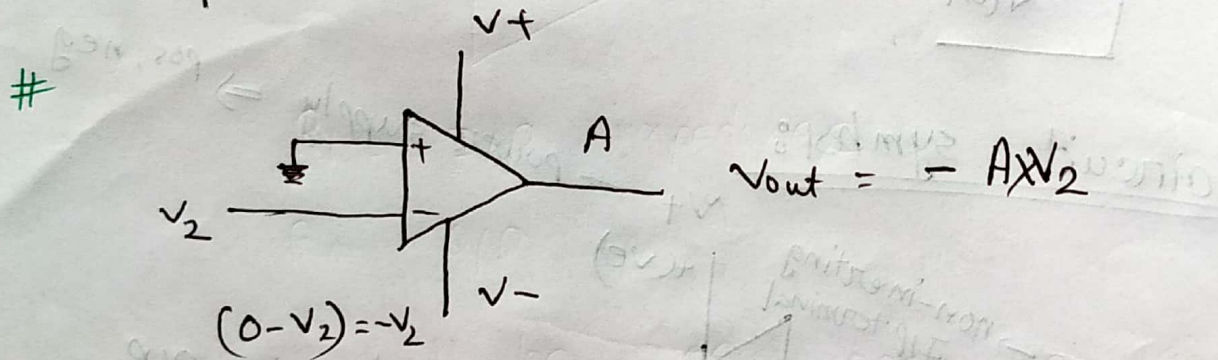
$$\text{then O/P} = A \times (V_1 - V_2)$$

Open loop configuration:



Here, A is the open loop gain - বহুগুন

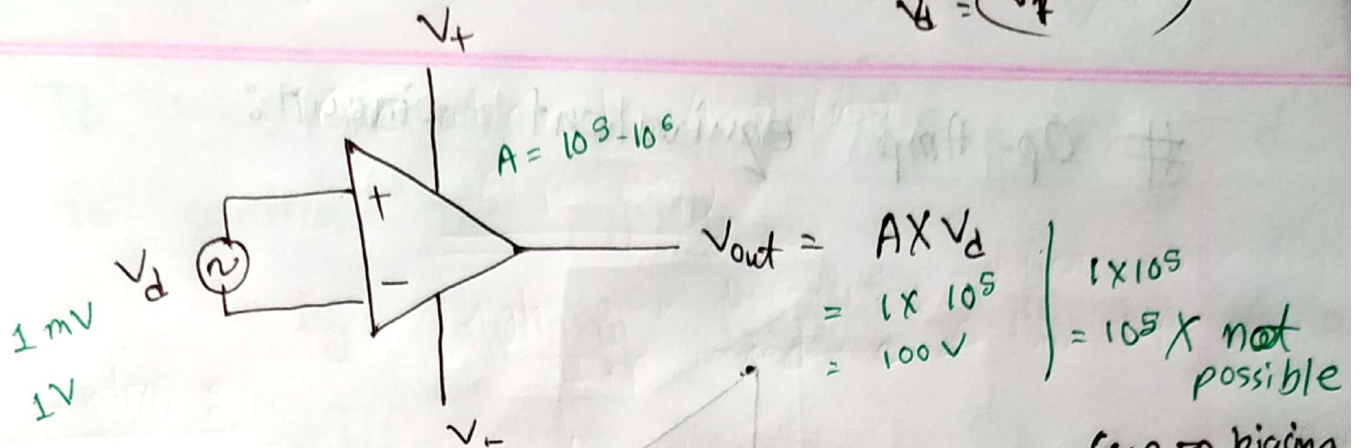
এখানে o/p থেকে I/P side-এ কোন feedback
নাহি,



o/p-এর signal has 180° phase with respect to I/P
Signal, so ~~non~~ inverting terminal.

$$V_d = (V_1 - V_2)$$

#

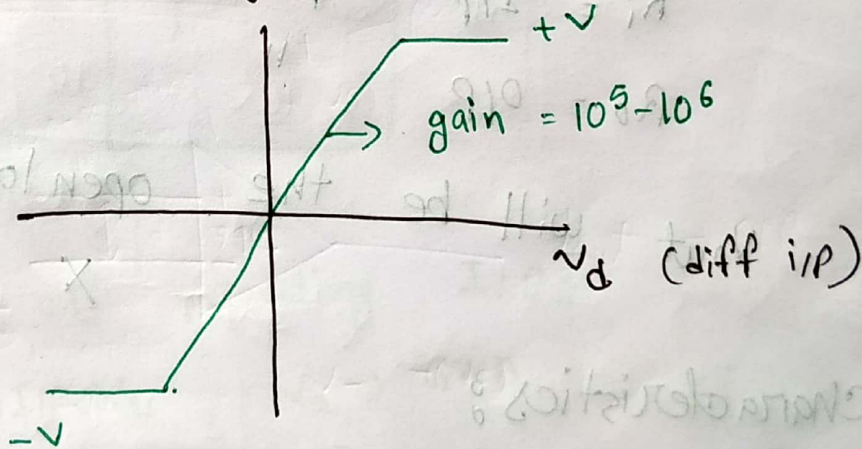


* Op-amp is high gain amp

Range, $A = 10^5 - 10^6$

Op-amp biasing
Voltage (V_{+} ,
restrict V_{-}
 A_{CL})

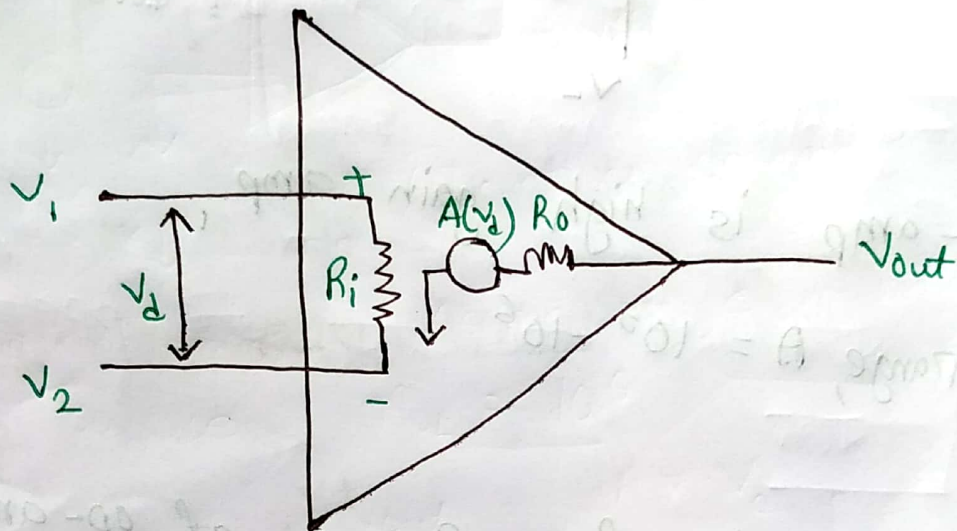
Voltage transfer curve of op-amp:



Applications:

- 1) Active filters
- 2) Oscillators
- 3) Waveform converter
- 4) ADC and DAC

Op-Amp equivalent circuit:



$R_i = \text{IIP impedance}$

$R_o = \text{OIP}$

V_{out} will be the open loop gain (A) $\times V_d$

Characteristics:

ideal

$$R_i = \infty$$

$$R_o = 0$$

$$\text{Bandwidth} = \infty$$

$$A = \infty$$

$$\text{IIP} = 0, \text{OIP} = 0$$

Practical

$$R_i \text{ range } M\Omega$$

$$R_o \text{ n } k\Omega$$

$$A \approx 10^5 - 10^6$$

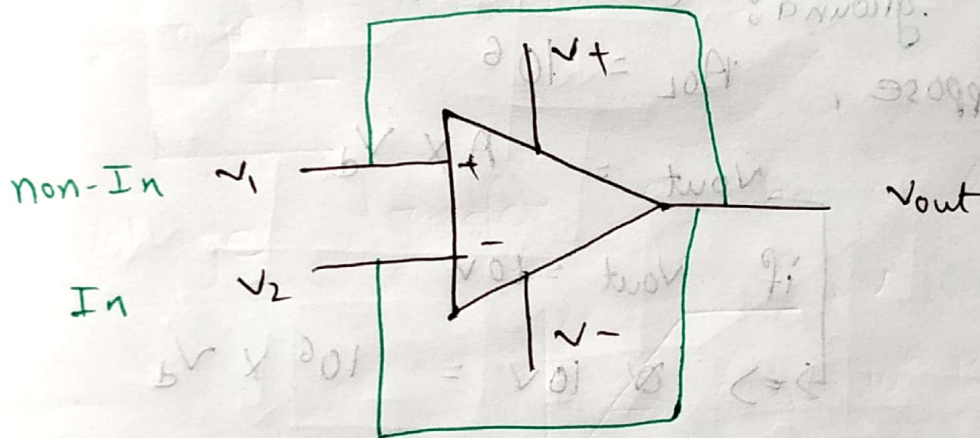
$$\text{IIP} = 0, \text{OIP} \text{ some } mV$$

To use op-amp as Amplifier we need to control the gain.

- by applying feedback from OP to I/P side

- 2 ways -

V_+ - Positive feedback
 V_- - negative feedback



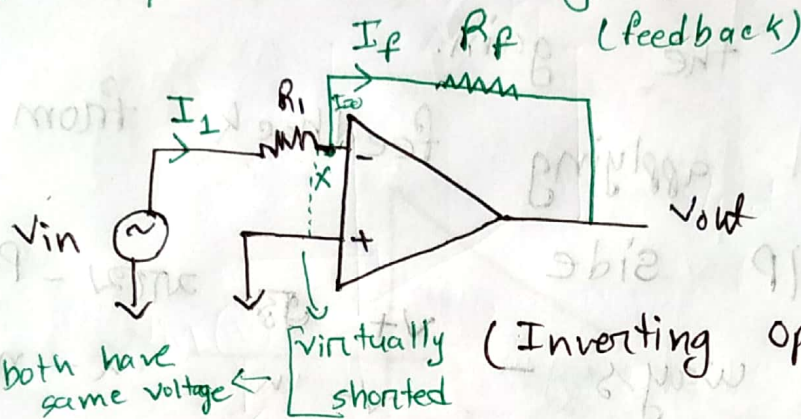
Ways of providing I/P:

1) I/P $\Rightarrow V_1$, $V_2 \Rightarrow \text{GND}$

2) I/P $\Rightarrow V_2$, $V_1 \Rightarrow \text{GND}$

3) I/P both V_1, V_2

I/P in inverting terminal :



(Inverting op-amp configuration)

* virtual ground: (same voltage in both terminal)

suppose, $A_{OL} = 10^6$

$$V_{out} = A \times V_d$$

if $V_{out} = 10V$

$$\Rightarrow 10V = 10^6 \times V_d$$

$$\Rightarrow V_d = 10\mu V$$

$$\Rightarrow V^+ - V^- = 10\mu V \quad [\text{small can neglect}]$$

$$\Rightarrow V^+ - V^- \approx 0V$$

$$\Rightarrow V^+ = V^- \quad [\text{they are in same potential}]$$

$$V^+ \Rightarrow \text{ground} = 0$$

$$V^- = 0$$

[virtual short]

virtually short circuited

Applying KCL at node X,

$R_{in} = \infty$, so no current will enter into op-amp
so $I = 0$

$$I = I_i - I_f$$

$$I_f - I_f = 0 \Rightarrow I_1 = I_f$$

$V_x =$ Voltage in x node

$$\frac{V_{in} - V_x}{R_1} = \frac{V_x - V_{out}}{R_f}$$

by using concept of virtual ground,

$$V_x = 0$$

$$\Rightarrow \frac{V_{in}}{R_1} = \frac{-V_{out}}{R_f}$$

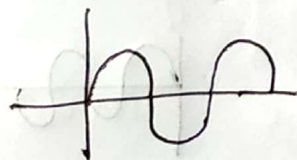
$$\Rightarrow \frac{V_{out}}{V_{in}} = - \frac{R_f}{R_1} = A_{CL}$$

180° neg phase

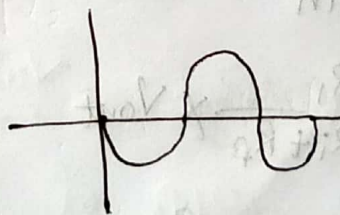
(close loop gain)

$$\Rightarrow V_{out} = - \frac{R_f}{R_1} \times V_{in}$$

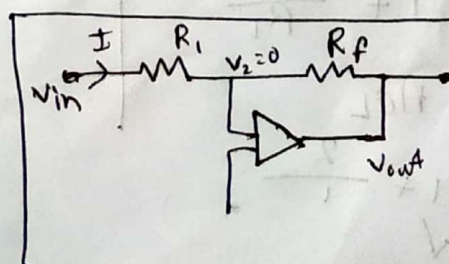
I/P



O/P



inverted



$$\text{Current, } I = \frac{V_{in} - V_{out}}{R_1 + R_f}$$

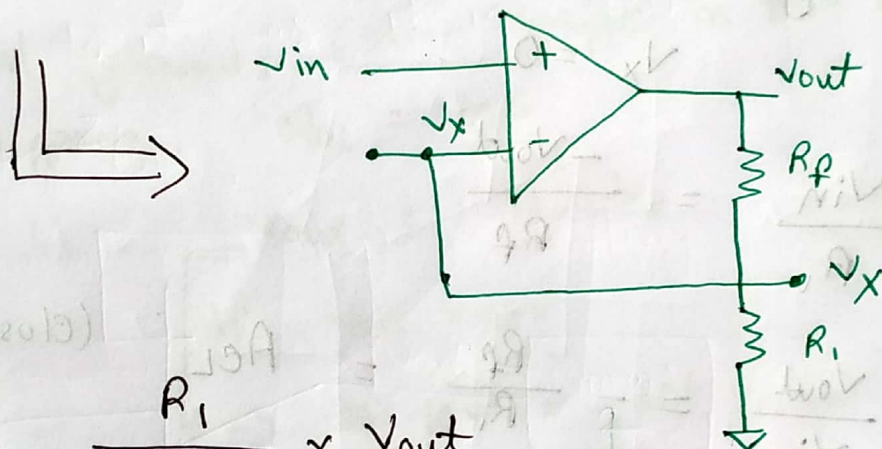
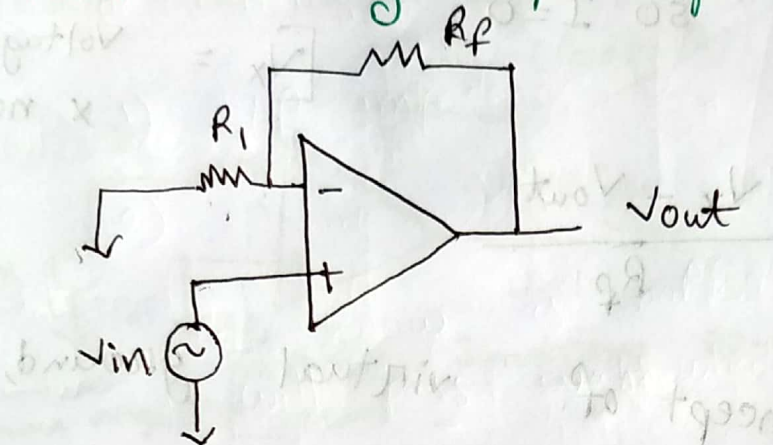
$$I = \frac{V_{in} - V_2}{R_1} = \frac{V_2 - V_{out}}{R_f}$$

$$I = \frac{V_{in}}{R_1} - \frac{V_2}{R_1}$$

$$= \frac{V_2}{R_f} - \frac{V_{out}}{R_f}$$

Non-inverting Op-Amp :

[Non-inverting op-Amp configuration]



as feedback
going to terminal

$$V_x = \frac{R_1}{R_1 + R_f} \times V_{out}$$

$$[V^+ = V^-] \text{ V. short}$$

$$V^+ = V_{in}$$

$$V^- = V_{in} = V_x$$

$$V_{in} = \frac{R_1}{R_1 + R_f} \times V_{out}$$

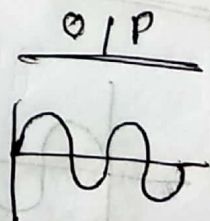
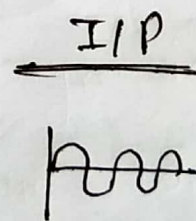
$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{R_1 + R_f}{R_1}$$

$$= 1 + \frac{R_f}{R_1}$$

$$= A_{CL}$$

$$A_{CL} = 1 + \frac{2}{1} = 3V$$

$$R_1 = 1, \\ R_f = 2$$



3V amplified signal

Advantages :

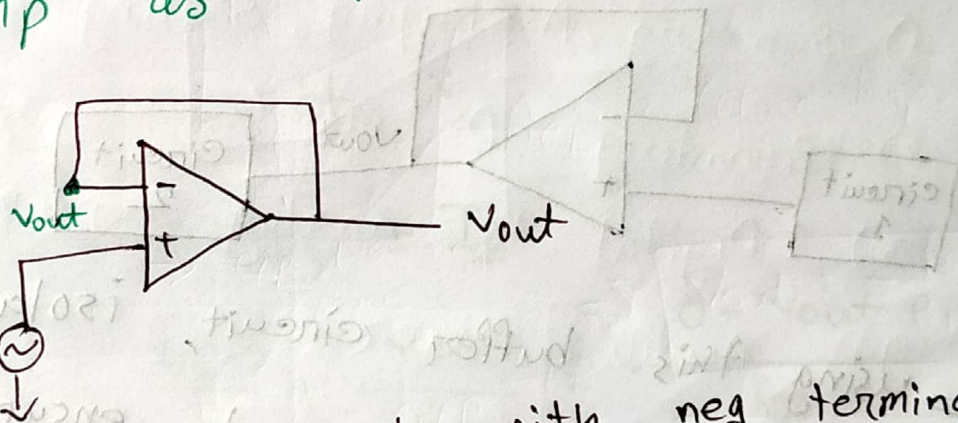
Non-inverting

- 1) same phase
- 2) I/P impedance very high

Inverting

- 1) 180° phase shift
- 2) depends of R_1

Op-Amp as buffer :



* V_{out} is shorted with neg terminal

$$V^+ = V^- \quad [\text{virtual short}]$$

$$V^- = V_{in} = V_{out}$$

* Op voltage follows the I/P voltage

* So it is called voltage-follower circuit

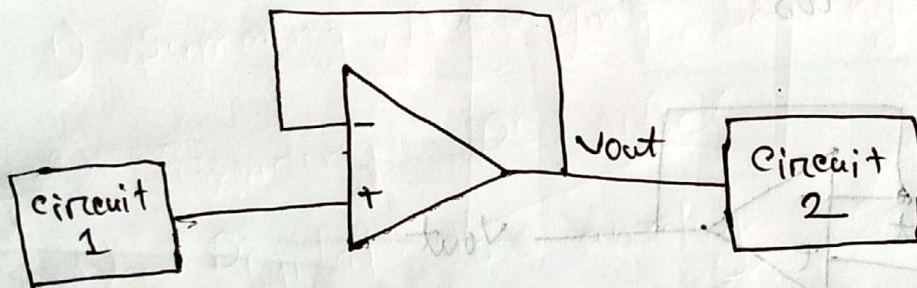
Characteristics :

1) I/P impedance, $Z_{in} = \infty$

2) $V_{out} = V_{in}$

- also known as buffer circuit. Because it will pass ~~with~~ whatever I/P coming to it as it is and provide high

Z_i

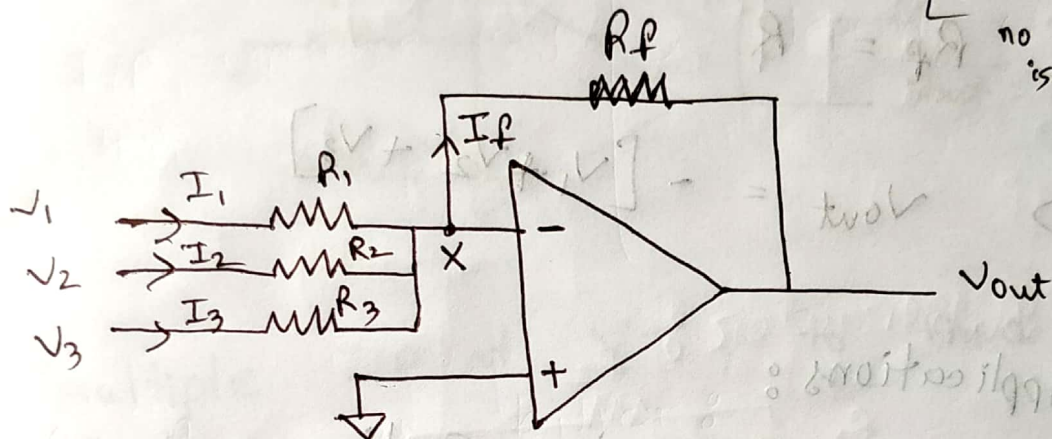


using this buffer circuit, isolate the 2 diff circuits and ensure that whatever the voltage is appearing at the o/p of the one circuit will appear at the I/P of another circuit.

Adder (Summing amplifier)

inverting

[ideal op-amp]
no current
is going into the
op-amp



Applying KCL at X,

$$I_1 + I_2 + I_3 = I_f$$

$$[V_X = 0]$$

$$\Rightarrow \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

single I/P voltage,

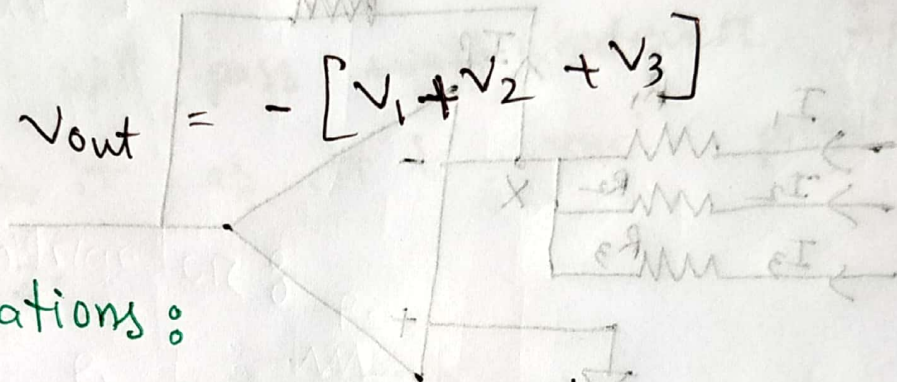
$$V_{out} = - \frac{R_f}{R_1} V_{in}$$

if $R_1 = R_2 = R_3 = R$

$$V_{out} = - \frac{R_f}{R} [V_1 + V_2 + V_3]$$

if $R_f = R$

$$\Rightarrow V_{out} = - [V_1 + V_2 + V_3]$$

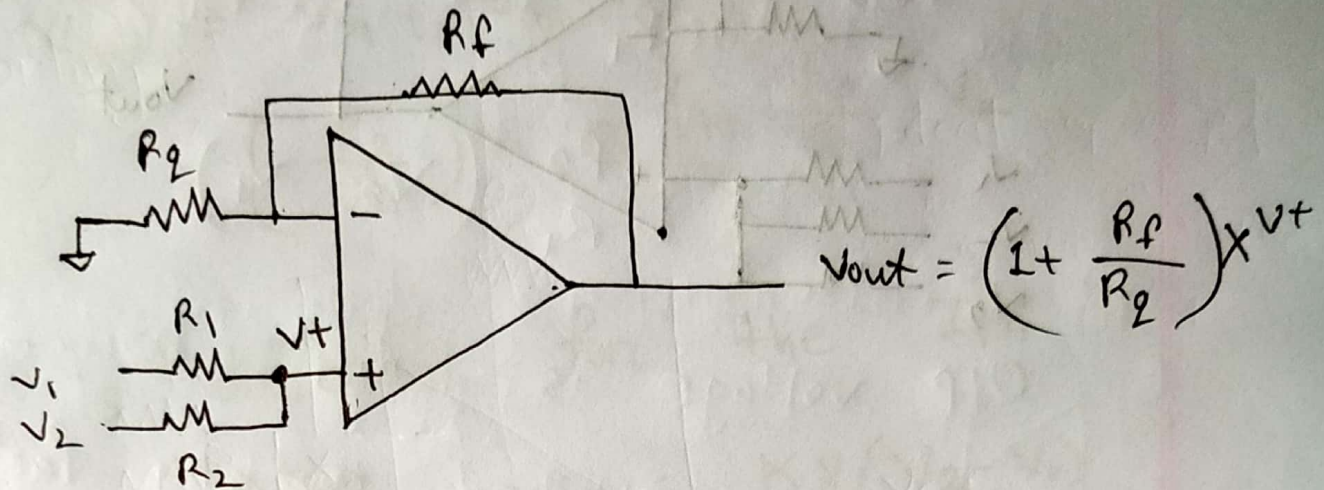


Applications :

- 1) Summing, Averaging, Scaling
- 2) Providing DC offset
- 3) Digital to analog converter
- 4) Audio mixer

non-inverting

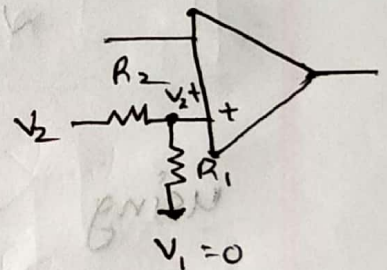
2 input



multiple I/P at V_{+} , to find V_{+} we use superposition theorem,

$$V_{2+} = \frac{R_1}{R_1 + R_2} \times V_2$$

same way, $V_{1+} = \frac{R_2}{R_1 + R_2} \times V_1$



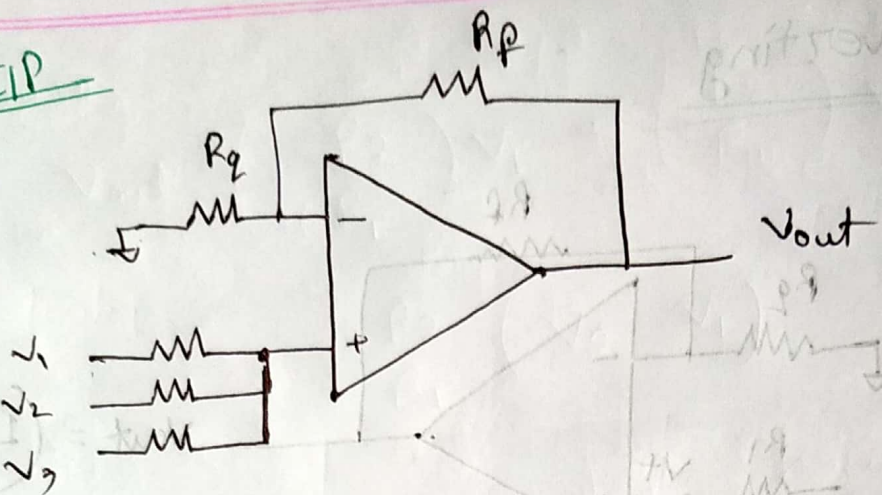
$$V_{+} = V_{1+} + V_{2+} = \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$$

$$V_{out} = \left(1 + \frac{R_f}{R_g}\right) \times V_{+} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2 \right)$$

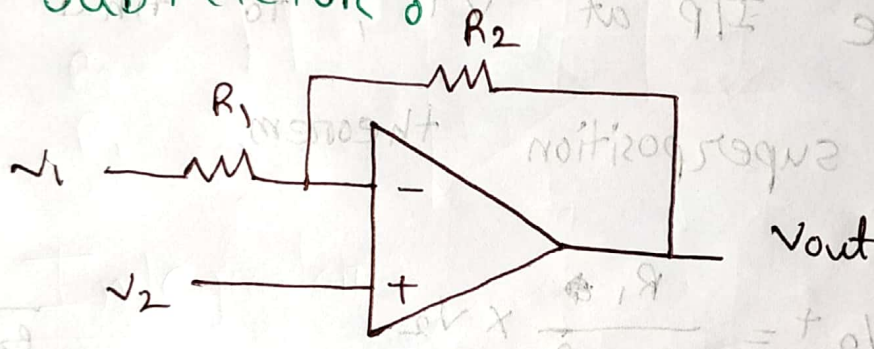
$$R_f = R_g = R \Rightarrow V_{out} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{V_1 + V_2}{2} \right)$$

$$R_f = R_g \Rightarrow V_{out} = V_1 + V_2$$

3 IIP



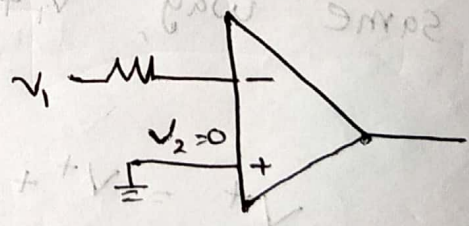
Subtractor



using superposition,

1) $V_2 = 0$

$$V_{01} = \left(-\frac{R_2}{R_1} \right) V_1$$



inverting configuration

same, $V_1 = 0$

$$V_{02} = \left(1 + \frac{R_2}{R_1} \right) V_2 \quad (\text{non-inverting})$$

3) V_1 & V_2 working simultaneously,

$$V_{out} = V_{o1} + V_{o2}$$

$$= \left(-\frac{R_2}{R_1} \right) V_1 + \left(1 + \frac{R_2}{R_1} \right) V_2$$

* Gain is different for the I/P

if gain is equal, $V_{out} = K \times (V_2 - V_1)$

so we need to make the term $\left(1 + \frac{R_2}{R_1} \right)$ equal to $\frac{R_2}{R_1}$

$$1 + \frac{R_2}{R_1}$$

$$= \frac{R_1 + R_2}{R_1}$$

$$= \frac{R_1 + R_2}{R_1} \times \frac{R_2}{R_1 + R_2} = \frac{R_2}{R_1}$$

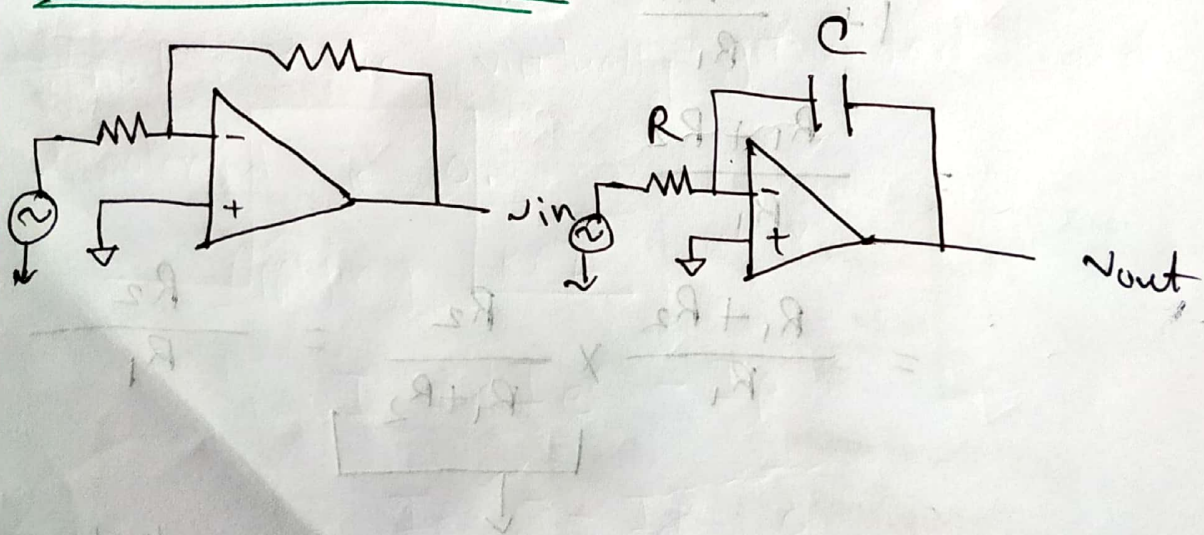
Here we apply voltage divider instead of applying the voltage source V_2 directly

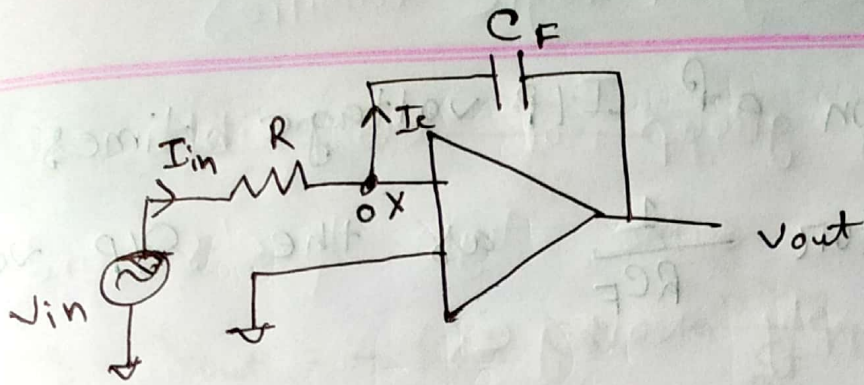
Now,

$$V_{out} = \left(\frac{-R_2}{R_1} \right) V_1 + \left(\frac{R_2}{R_1} \right) \times V_2$$
$$= \frac{R_2}{R_1} (V_2 - V_1)$$

OP voltage is diff between two
voltage multiplied ~~with~~ by factor of
 $\frac{R_2}{R_1}$

Integrator:





Applying KCL, at X,

$$I = I_{in} - I_c$$

$$\Rightarrow I_{in} - I_c = 0$$

$$\Rightarrow I_{in} = I_c$$

$$\Rightarrow \frac{V_{in} - 0}{R} = I_c$$

$$\Rightarrow \frac{V_{in}}{R} = C_F \times \frac{dV_c}{dt}$$

$$= C_F \times \frac{d}{dt} [0 - V_{out}]$$

$$\Rightarrow \frac{V_{in}}{R} = - C_F \times \frac{d}{dt} V_{out}$$

$$\Rightarrow \frac{dV_{out}}{dt} = - \frac{1}{RC_F} \times V_{in}(t)$$

integrating both sides,

$$V_{out}(t) = - \frac{1}{RC_F} \int V_{in}(t) dt$$

[because of feedback, X is virtually grounded]
potential = 0

capacitor current,

$$I_c = C \frac{dV_c}{dt}$$

$$Q = CV_c$$

$$I_c = \frac{dQ}{dt} = C \frac{dV_c}{dt}$$

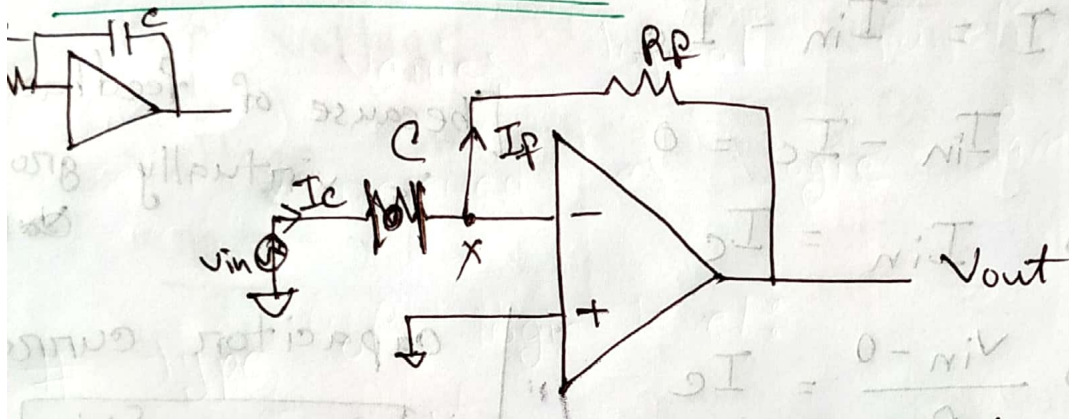
(Not changing with time)

we get,

integration of I/P voltage times

the factor $\frac{1}{RC_F}$ as the O/P voltage

Differentiator: differentiation of I/P



because of negative feedback the node X is virtually grounded and potential is 0.

Applying KCL at X,

$$I = I_c - I_f$$

$$\Rightarrow I_c - I_f = 0$$

$$\Rightarrow I_c = I_f$$

$$\Rightarrow \frac{C \frac{dV_c}{dt}}{1} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow C \frac{dV_{in}}{dt} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - R_f \times C \times \frac{dV_{in}}{dt}$$

in this circuit, O/P voltage is the differentiation of the I/P voltage times factor $R_f C$

block diagram of op-amp in sensor based data acquisition:

